

Towards Foundation Models for Quantum Unitary Synthesis via Zero-Shot MDL

Lukas Thei inger^{1,3} Thore Gerlach² David Berghaus^{1,4}
¹Lamarr Institute ²European Space Agency ³University of Bonn ⁴Fraunhofer IAIS

Problem setting

- Given: Target unitary U
- Goal: Recover an exact circuit over the Clifford+T gate set
- Search: Append one gate at a time until the residual unitary is solved

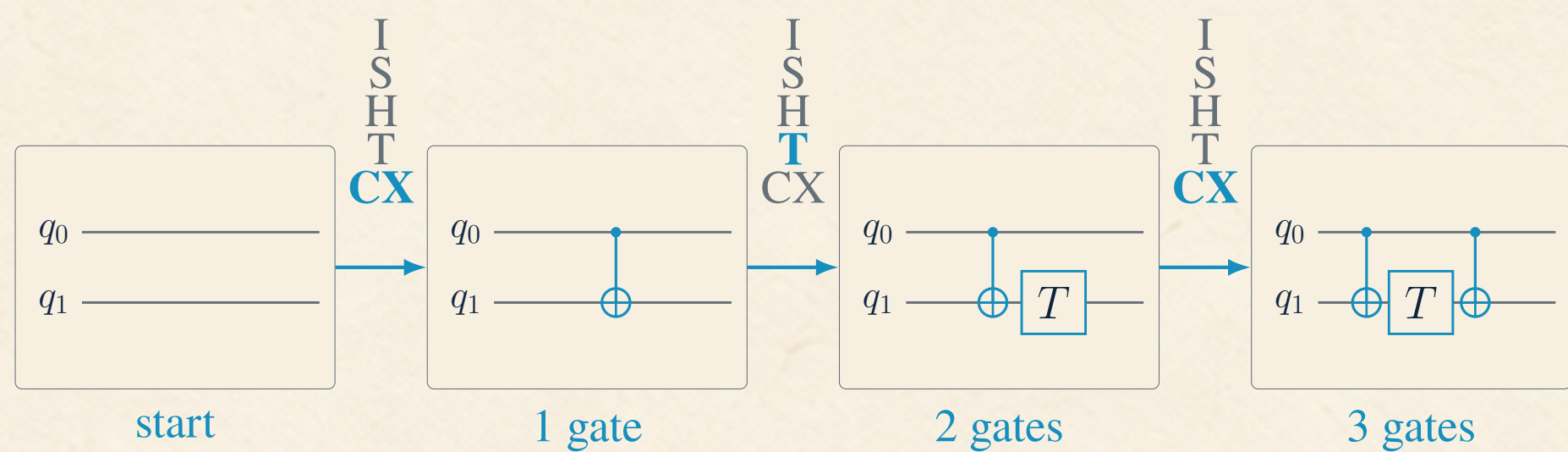


Figure 1. Example synthesis task for $U = \text{diag}(1, e^{i\pi/4}, e^{i\pi/4}, 1)$: recover the exact 3-gate circuit $CX_{01}T_1CX_{01}$. Blue labels show the selected gate at each step

Challenges

- The problem is NP-hard in general
- Search space grows exponentially with the length of the circuits
- Standard unitary distances do not provide reliable search guidance

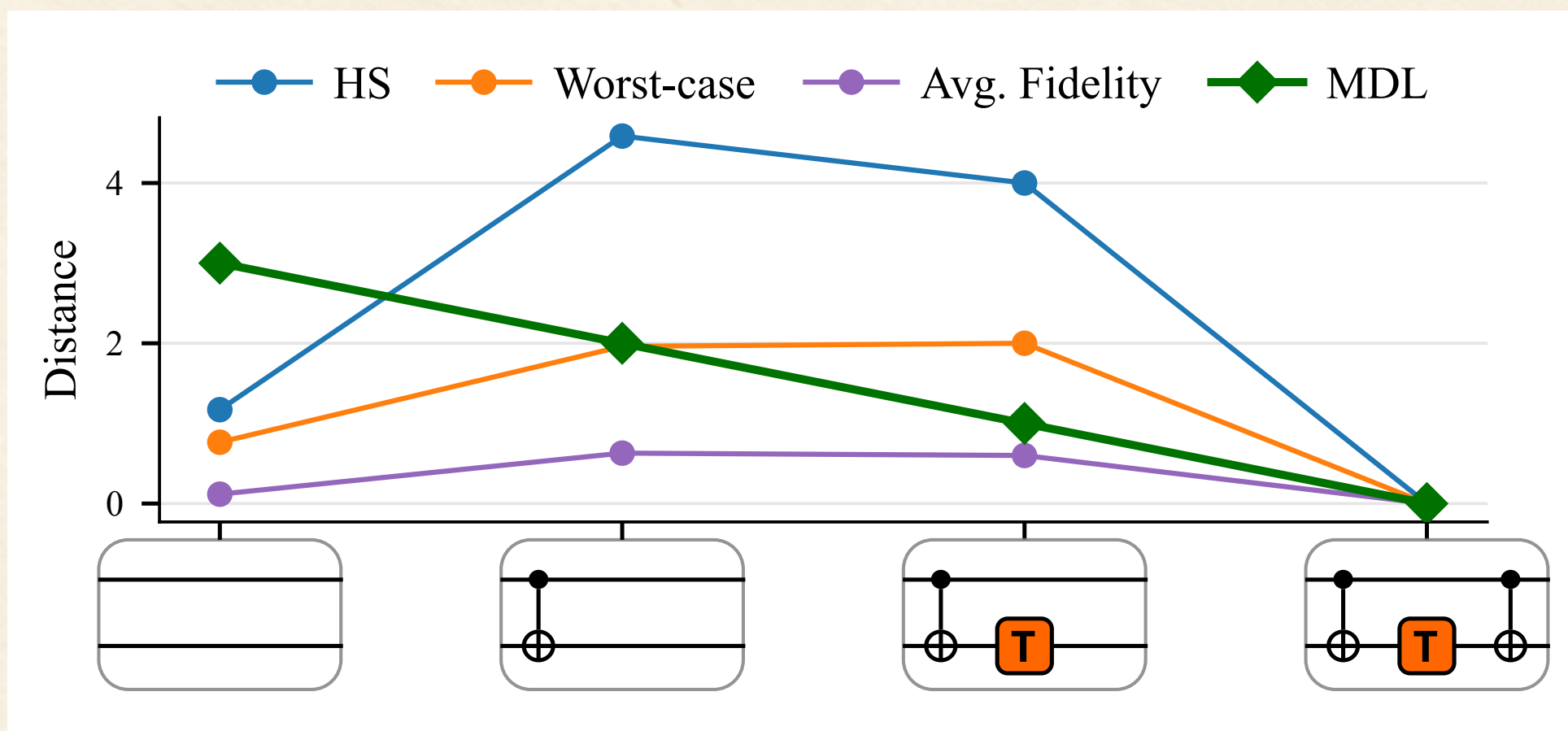


Figure 2. Along the correct path, Hilbert-Schmidt, worst-case and average-fidelity error are non-monotone, while MDL drops cleanly from 3 to 0.

Minimum description length (MDL)

- MDL principle:** Prefer the shortest exact explanation
- For synthesis:** MDL is the remaining gate count of the shortest exact circuit
- Search view:** Along a correct path, MDL decreases by one per step
- Challenge:** Exact MDL is NP-hard, so we learn a predictor instead

Training MDL Predictor

- Generate circuits** Sample Clifford+T circuits on the fly and rebalance the dataset across T -counts
- Build labels** Run a fast optimizer; the optimized suffix length becomes the approximate MDL target. For harder targets, also include easier partial circuits
- Train predictors** Phase-normalize the unitary and train either an MLP on matrix entries or a Transformer on tokenized inputs
- Loss** Regress the scalar MDL target with MSE

Inference: Stochastic Beam Search

- Couple MDL Predictor with Beam search
- Rank partial circuits by predicted remaining MDL
- Beam search is fast and highly parallelizable
- Add stochasticity and equivalent root states to improve exploration

Beam search view

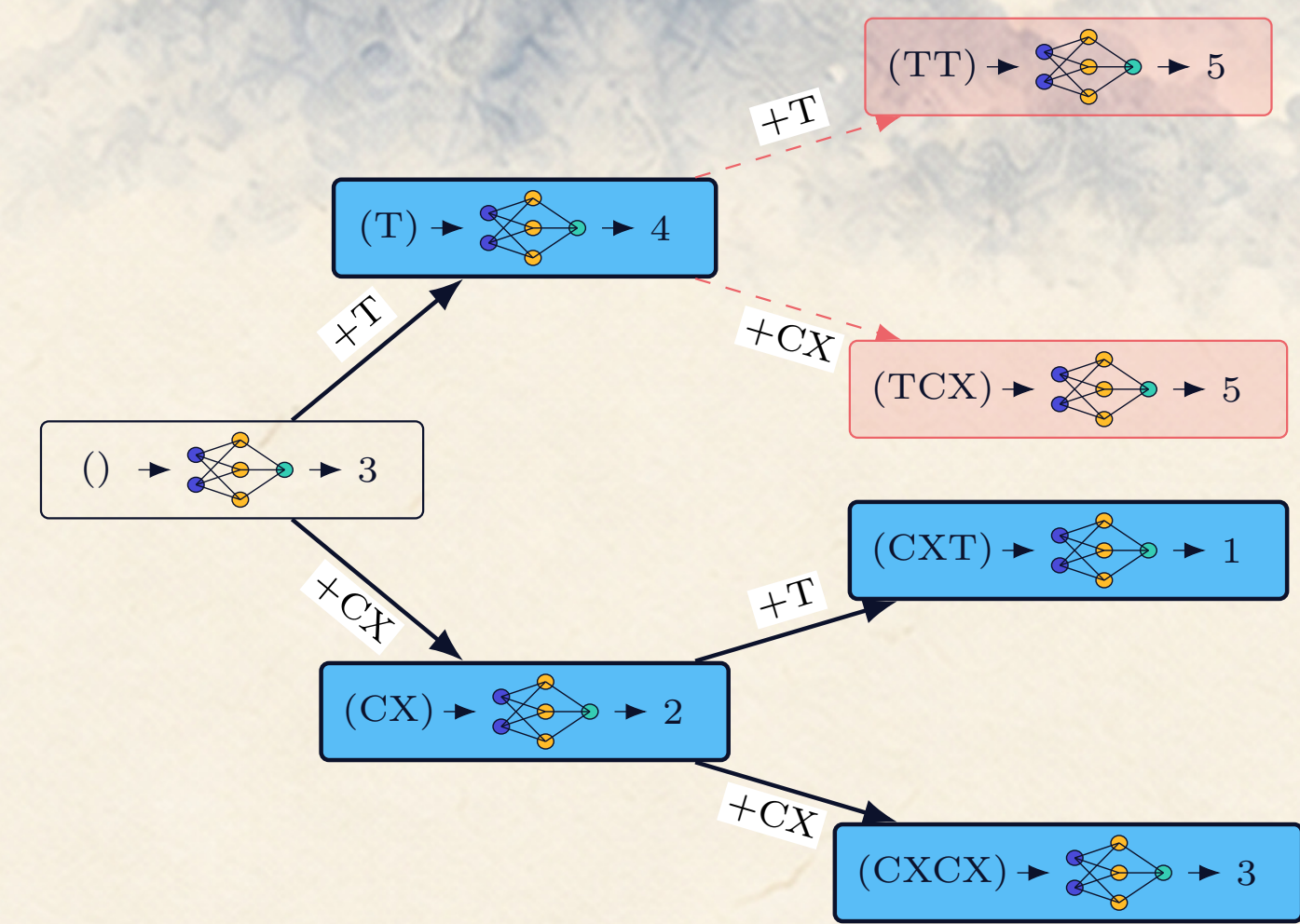


Figure 3. Beam-search view: candidate partial circuits are expanded, ranked by predicted remaining MDL, and pruned back to the beam

Takeaways

A simple and reusable, **zero-shot** MDL-guided search heuristic for quantum unitary synthesis that is cheap to train and fast at inference

6 h vs. 7 d

MLP training cost versus
the reported RL baseline

22 s

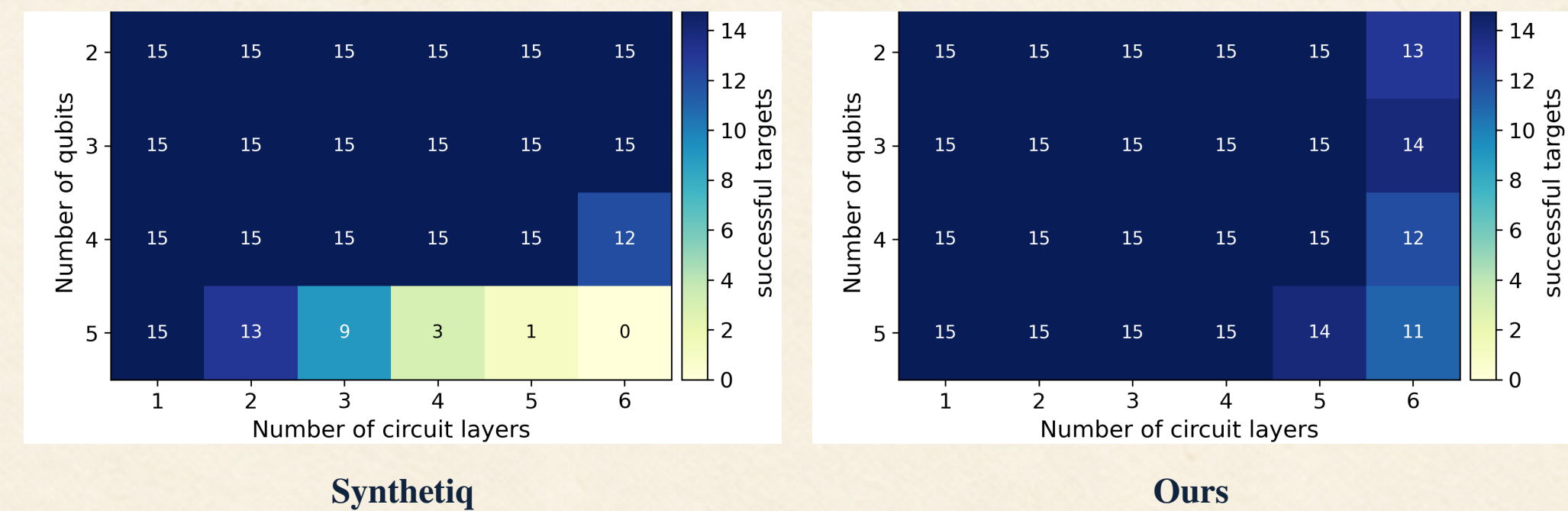
average runtime per 5-qubit
instance at test time

15/15

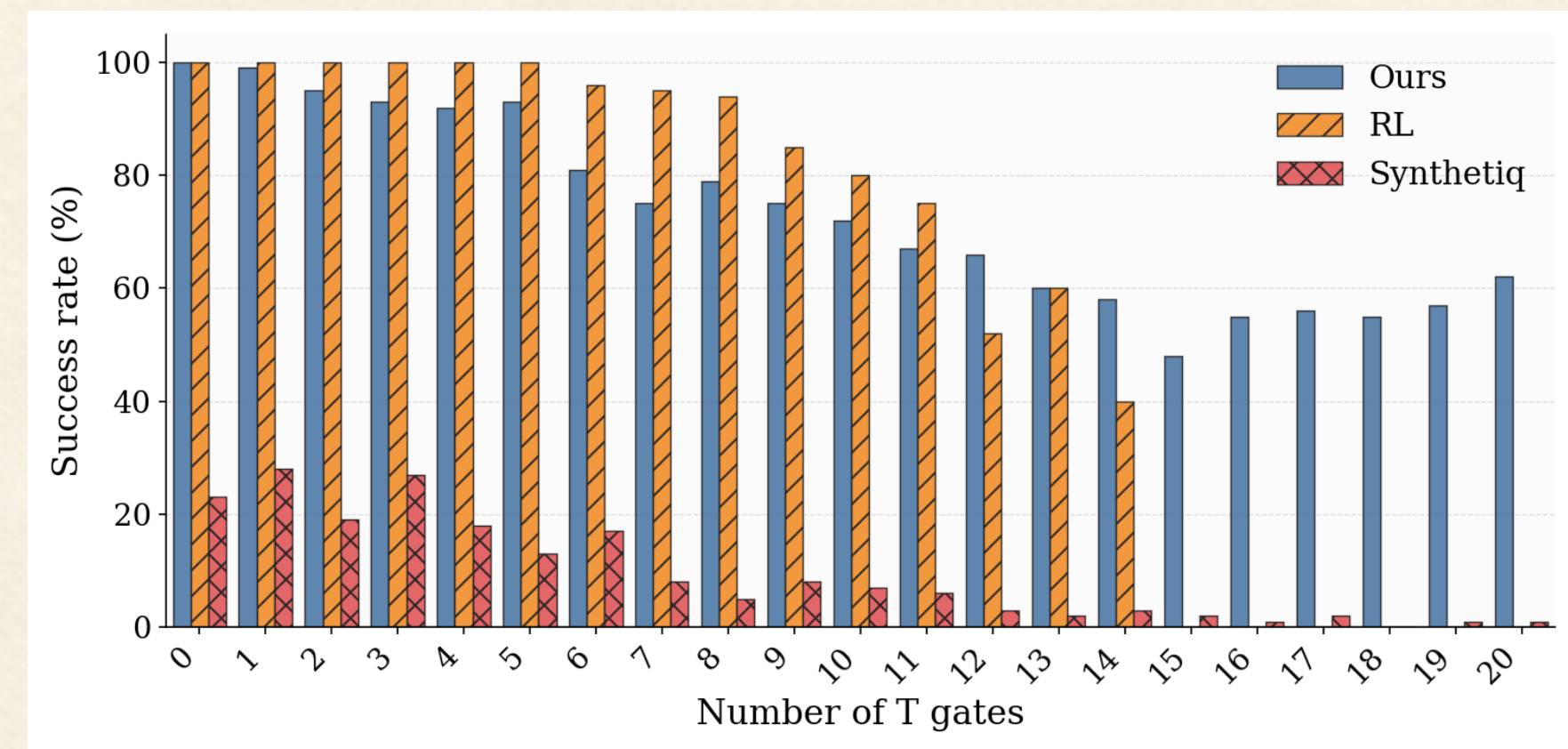
successes in almost every
QAS-Bench bucket

Results

QAS-Bench standardized Benchmark. Testing on 2–5 qubits, difficulty levels 1–6 with 15 targets per bucket. We solve nearly every bucket 15/15 and beat Synthetiq on the hard multi-qubit cases.



Random targets Success degrades more slowly as the T-count increases in comparison to RL.



RL bars are only approximate. Additional comparisons can be found in the paper

Limitations

- Search states are dense $2^n \times 2^n$ matrices, so memory still scales as $\Theta(4^n)$
- Performance depends on the training distribution
- Beam search trades compute for quality and has no worst-case guarantee
- Cannot be used directly with parametric gate sets